Lecture 6 - The Derivative as a Function

Garrett Credi

09/24/2026

Quiz Time!

Please get into organized rows so that you have a “bubble” of space around you away from other people.
Reminder:

1. Do not start the quiz until I announce you can start.
2. Do not read off other students’ work to cheat
3. Place all class material (notes, worksheets, etc) and devices (phones, tablets, headphones, etc) away
4. Don’t cheat and or be shady... obviously...

Last Time

Last lecture we:

1. Introduced the idea of linearly approximating functions as an improvement to the constant-value approximation $f(x) \approx f(a)$ for $x \approx a$ of continuous functions.
2. Defined the *derivative* of a function $f(x)$ at a point $x = a$ as

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{1}$$

3. Saw that this quantity was also:
   (1) The slope of the best linear approximation

$$\mathscr{L}_{x=a}(x) = f(a) + f'(a)(x - a) \tag{2}$$

   to $f(x)$ at $x = a$.
   (2) The slope of the tangent line to the graph of $y = f(x)$ at $(a, f(a))$.
   (3) The instantaneous velocity of a particle whose motion is described by $f(x)$ at $x = a$.

Last Time Cont.

We also were able to compute some derivatives using strategies from our limit section of the course e.g. for $f(x) = x^2$:

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \quad (3)$$

$$= \lim_{x \to 0} \frac{x^2}{x} \quad (4)$$

$$= \lim_{x \to 0} x = 0 \quad (5)$$

$$f'(1) = \lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} \quad (6)$$

$$= \lim_{x \to 1} \frac{x^2 - 1}{x - 1} \quad (7)$$

$$= \lim_{x \to 1} x + 1 = 1 + 1 = 2 \quad (8)$$

$$f'(2) = \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2} \quad (9)$$

$$= \lim_{x \to 2} \frac{x^2 - 4}{x - 2} \quad (10)$$

(11)

Wait... these all seem pretty similar!

The Derivative as a Function

The Derivative Function I

What kind of mathematical object do we use when we want to compactly describe the process of taking some input value a and computing an output quantity $f'(a)$?
A function!
Let's define *a derivative function* to see if we can simplify and speed up computations:

The Derivative Function

For a function $f(x)$ we define the *derivative* (function), denoted either as:

$$f'(x) \text{ or } \frac{df}{dx} \text{ or } \frac{d}{dx}f(x),$$

to be the function so that:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{u \to x} \frac{f(u) - f(x)}{u - x} \tag{12}$$

Is this different from what we were doing before?

The Derivative Function II

Yes! We can now compute *every* value of a derivative at once! E.g. for $f(x) = x^2$ again

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{13}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \tag{14}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \tag{15}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h} \tag{16}$$

$$= \lim_{h \to 0} 2x + h = 2x + 0 = 2x \tag{17}$$

So we immediately get that:

1. $f'(0) = 2 \cdot 0 = 0$
2. $f'(1) = 2 \cdot 1 = 2$
3. $f'(2) = 2 \cdot 2 = 4$

The Derivative Function III

Big Idea of Making it a Function

Here're two very important reasons why we might want to treat the derivative as a function and not just compute value by value:

1. Speed. Once we do the (slightly length) work of computing the general functional form of the derivative all future derivatives at a point can be computed *so quickly!* just by evaluation!
2. Information Extraction. Without the equation of the function, it's *impossible* to analytically study interesting properties of the derivative like:
   (1) For which values is $f'(a)$ positive? (when is $f(x)$ increasing?)
   (2) For which values is $f'(a)$ zero? (important for maximizing/minimizing which we'll do in 3/4 weeks)
   (3) For which values is $f'(a)$ undefined? (when can we not linearly approximate?)

   but given an equation of a function it's fairly easy to answer all of those!

The Derivative Function IV

For example: at which x values is the slope of the line tangent to $f(x) = x^3 - 2x + 1$ equal to zero?

Step 1: Translate to derivative question: when is $f'(x) = 0$?

Step 2: Compute derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{18}$$

$$= \lim_{h \to 0} \frac{(x+h)^3 - 2(x+h) + 1 - (x^3 - 2x + 1)}{h} \tag{19}$$

$$= \lim_{h \to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h + 1 - x^3 + 2x - 1}{h} \tag{20}$$

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - 2h}{h} \tag{21}$$

$$= \lim_{h \to 0} 3x^2 + 3xh + h^2 - 2 \tag{22}$$

$$= 3x^2 + 3x \cdot 0 + 0^2 - 2 = 3x^2 - 2 \tag{23}$$

The Derivative Function V

Last slide: $f'(x) = 3x^2 - 2$.
Step 3: Answer derivative question: when is $f'(x) = 3x^2 - 2 = 0$?

$$3x^2 - 2 = 0 \tag{24}$$
$$3x^2 = 2 \tag{25}$$
$$x^2 = \frac{2}{3} \tag{26}$$
$$x = \pm\sqrt{\frac{2}{3}} \tag{27}$$

Step 4: Translate derivative question answer to original question answer:

When x is $\pm\sqrt{\frac{2}{3}}$, the slope of the line tangent to $f(x) = x^3 - 2x + 1$ is equal to zero.

The Derivative Function VI

This is the general strategy that you can use to solve *sooo* many questions about the behavior of functions!
We'll see that in the worksheet for today :)

The Derivative Function VII

Some conceptual questions:

1. For some random function $f(x)$ whose *domain* is all real numbers $\mathbb{R}$ must it also be the case that the domain of $f'(x)$ is also $\mathbb{R}$?
2. How do the following conditions relate:

(1) $f(x)$ is defined at $x = a$
(2) $\lim_{x \to a} f(x)$ exists
(3) $f(a) = \lim_{x \to a} f(x)$
(4) The domain of $f(x)$ is all real numbers $\mathbb{R}$
(5) $\lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}$ exists
(6) $f'(a)$ exists
(7) $f'(x)$ is defined at $x = a$
(8) $f(x)$ is differentiable at $x = a$
(9) $f'(x)$ is continuous at $x = a$
(10) $f'(x)$ is differentiable at $x = a$
(11) The domain of $f'(x)$ is all real numbers $\mathbb{R}$

3. Which of the above conditions relied on $f'(x)$ *being a function* rather than being a limit defined at a point $x = a$?

Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class!

That’s all folks :)